

Suggested Solutions to Exercises 4

1. **Put-Call Parity.** This exercise presents an important relationship between the price of a European call option and the price of a European put option on a same underlying non-dividend paying stock, with same strike price X and maturity date T . To derive the relationship we consider the following two portfolios

- *Port. A:* Long 1 call option + Xe^{-rT} cash;
- *Port. B:* Long 1 put option + 1 share,

where r is the risk-free interest rate. If we denote, as usual, the stock price at T by S_T , then the value of both portfolios at time T is

$$\max(S_T, X)$$

for any value of S_T . In fact, since the options are European, they can only be exercised on the expiration date T , where we can only have the two following cases:

- If $S_T > X$:
 - * *Port. A:* My cash is now equal to X , and it is convenient to exercise the call option, hence I use X to buy the share, which is now worth $S_T \Rightarrow$ the portfolio is now worth S_T .
 - * *Port. B:* It is not convenient to exercise the put option, so I just keep my share, which is worth $S_T \Rightarrow$ the portfolio is now worth S_T .
- If $S_T < X$:
 - * *Port. A:* My cash is now equal to X , and it is not convenient to exercise the call option, hence I keep $X \Rightarrow$ the portfolio is now worth X .

- * *Port. B*: Now it is convenient to exercise the put option, so I sell the share for $X \Rightarrow$ the portfolio is now worth X .

We have just shown that the two portfolios have always the same value at T , whatever the value of S_T . It follows that they must also have identical values at time 0. Such values are

- *Port. A*: Call + Xe^{-rT}
- *Port. B*: Put + S

where S is the value of the stock at time 0. If we equate them, we obtain:

$$\text{Call} - \text{Put} = S - Xe^{-rT}$$

The two values are the same when $X = Se^{rT}$, i.e. when the strike price equals the future expected value of the stock in a risk-neutral world, or the fair value of a forward contract struck at time t .

2. Consider the following two portfolios:

- *Port. A*: Long 1 binary call + long 1 binary put
- *Port. B*: e^{-rT} cash

The payoff at time T from both portfolios is equal to £1. In fact

- If $S_T > X$:
 - * *Port. A*: Exercise the call and get £1. Do not exercise the put (get 0) $\Rightarrow$ total payoff £1.
 - * *Port. B*: My cash is now equal to 1.
- If $S_T < X$:
 - * *Port. A*: Do not exercise the call (get 0). Exercise the put and get £1 $\Rightarrow$ total payoff £1.
 - * *Port. B*: My cash is now equal to 1.

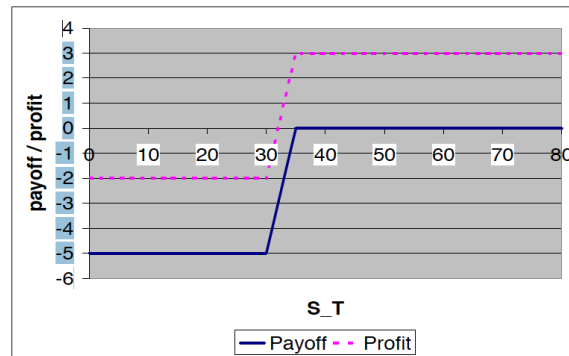
Since the two portfolios have the same payoff at T , their value at time 0 must be the same, i.e.

$$\text{Binary Call} + \text{Binary Put} = e^{-rT}$$

3. A bull spread is created by buying the \$30 put and selling the \$35 put. This strategy gives rise to an initial cash inflow of \$3. The outcome is as follows:

Stock Price	Payoff	Profit
$S_T \geq 35$	0	3
$30 \leq S_T < 35$	$S_T - 35$	$S_T - 32$
$S_T < 30$	-5	-2

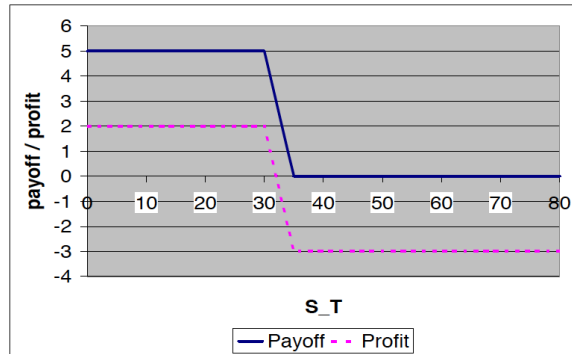
The payoff and profit functions are shown in the following diagram:



A bear spread is created by selling the \$30 put and buying the \$35 put. This strategy costs \$3 initially. The outcome is as follows:

Stock Price	Payoff	Profit
$S_T \geq 35$	0	-3
$30 \leq S_T < 35$	$35 - S_T$	$32 - S_T$
$S_T < 30$	5	2

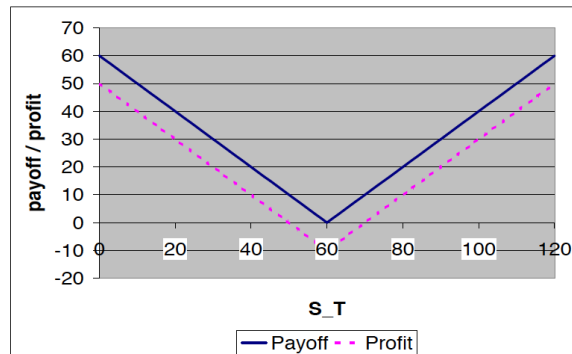
The payoff and profit functions are shown in the following diagram:



4. A straddle is created by buying both the call and the put. This strategy costs \$10. The outcome is as follows:

Stock Price	Payoff	Profit
$S_T \geq 60$	$S_T - 60$	$S_T - 70$
$S_T \leq 60$	$60 - S_T$	$50 - S_T$

This shows that the straddle will lead to a loss if the final stock price is between \$50 and \$70, as it is also evident from the following diagram:



5. Consider a portfolio that is long one call with strike price X_1 , long one call with strike price X_3 , and short two calls with strike price X_2 (all European). The value of the portfolio at time T can be worked out in

four different situations:

$$S_T \leq X_1 : \text{ Portfolio value} = 0$$

$$X_1 < S_T \leq X_2 : \text{ Portfolio value} = S_T - X_1 > 0$$

$$\begin{aligned} X_2 < S_T \leq X_3 : \text{ Portfolio value} &= S_T - X_1 - 2(S_T - X_2) \\ &= X_2 - X_1 - (S_T - X_2) \geq 0 \end{aligned}$$

$$\begin{aligned} S_T > X_3 : \text{ Portfolio value} &= S_T - X_1 - 2(S_T - X_2) + S_T - X_3 \\ &= X_2 - X_1 - (X_3 - X_2) = 0 \end{aligned}$$

This shows that the value of the portfolio is always either zero or positive at the expiration of the options. In the absence of arbitrage possibilities it must be positive or zero at time 0 too. This means that $c_1 + c_3 - 2c_2 \geq 0$, i.e.

$$c_2 \leq 0.5(c_1 + c_3)$$

6. The solution is analogous to the previous one. Here we consider a portfolio that is long one put with strike price X_1 , long one put with strike price X_3 , and short two puts with strike price X_2 (all European). The value of the portfolio at time T is:

$$\begin{aligned} S_T \leq X_1 : \text{ Portfolio value} &= X_1 - S_T - 2(X_2 - S_T) + X_3 - S_T \\ &= X_3 - X_2 - (X_2 - X_1) = 0 \end{aligned}$$

$$\begin{aligned} X_1 < S_T \leq X_2 : \text{ Portfolio value} &= X_3 - S_T - 2(X_2 - S_T) \\ &= X_3 - X_2 - (X_2 - S_T) \geq 0 \end{aligned}$$

$$X_2 < S_T \leq X_3 : \text{ Portfolio value} = X_3 - S_T > 0$$

$$S_T > X_3 : \text{ Portfolio value} = 0$$

Since the portfolio value is always non-negative at the expiration, the same must hold at time zero. Hence $p_1 + p_3 - 2p_2 \geq 0$, i.e.

$$p_2 \leq 0.5(p_1 + p_3)$$